THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rectangle's length is 2 times its breadth and its perimeter is 54 cm. Its area (cm^2) is: 1

(a) 153 **(b)** 180 **(c)** 162 **(d)** 171

2. A square and an equilateral triangle have equal perimeters. If the square's side is 12 cm, the triangle's side (cm) is: 1

(a) 16 **(b)** 20 **(c)** 28 **(d)** 12

3. A rectangular field 24 m by 10 m is divided by a diagonal into two triangles. The area of one triangle (m^2) is: 1

(a) 144 **(b)** 96 **(c)** 130 **(d)** 120

4. A picture 30 cm by 20 cm is mounted on a frame of width 3 cm all around. The area of the frame alone (cm^2) is: 1

(a) 348 **(b)** 336 **(c)** 339 **(d)** 333

5. A wire in the shape of a square of side 20 cm is straightened and bent into a rectangle of length 26 cm. The breadth (cm) is: 1

(a) 26 **(b)** 6 **(c)** 14 **(d)** 8

6. Tiling a rectangular floor of length 9 m at ₹150 per m^2 costs ₹6750. Its breadth (m) is: 1

(a) 10 **(b)** 45 **(c)** 14 **(d)** 5

7. Out of all rectangles with perimeter 32 cm having whole-number sides, the greatest area (cm^2) is: 1

(a) 80 **(b)** 64 **(c)** 72 **(d)** 56

8. The sides of a triangle are 13 cm, 14 cm and 15 cm. Its perimeter (cm) is: 1

(a) 43 **(b)** 42 **(c)** 55 **(d)** 41

9. **Assertion (A):** The perimeter of a rectangle 12 m long and 5 m wide is 34 m. 1
Reason (R): Perimeter of a rectangle = 2 x (length + breadth).
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The area of a square of side 7 cm is 49 cm^2. 1
Reason (R): Area is measured in square units.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The length and breadth of a rectangle are in the ratio 3:2 and its perimeter is 60 cm. Find the sides. 2

12. A rectangular field is 40 m by 30 m. Find the cost of levelling the field at ₹5 per m^2 and fencing at ₹12 per metre. 2

13. A carpet 5 m by 4 m is laid in a room 8 m by 6 m. Find the area of the uncovered floor. 2

14. The length and breadth of a rectangle are in the ratio 5:3 and its perimeter is 64 cm. Find the sides. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangular plot 40 m by 25 m is to be fenced with 3 rounds of barbed wire costing ₹12 per metre. Find the total cost. Also find the cost of ploughing it at ₹4 per m^2. 3

16. A square of side 12 cm and a rectangle of length 14 cm and breadth 10 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

17. A floor 6 m by 4 m is to be covered with tiles each 30 cm by 20 cm. Find the number of tiles. 3

OR

A rectangular plot 50 m by 30 m is to be fenced with 2 rounds of barbed wire costing ₹15 per metre. Find the total cost. Also find the cost of ploughing it at ₹5 per m^2. 3

18. A square of side 15 cm and a rectangle of length 18 cm and breadth 12 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

OR

A floor 8 m by 5 m is to be covered with tiles each 40 cm by 25 cm. Find the number of tiles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Farmer Ramesh has a rectangular field 40 m long and 25 m wide. He will fence it with wire and then sow seeds. 5

(i) Find the perimeter of the field.

(ii) Find the area of the field.

(iii) Find the cost of fencing at ₹20 per metre.

(iv) 2 kg of seeds are needed for every 50 m^2. How many kg of seeds does he need?

20. A rectangle with length 24 cm and breadth 10 cm has a triangular piece with base 24 cm and height 6 cm cut from one side. Find the area of the remaining piece and the fraction of the whole that is left. 5

OR

A rectangular park 80 m by 50 m has two paths each 4 m wide crossing in the middle, one parallel to each side. Find the area of the paths and the area of the rest of the park. 5

END OF QUESTION PAPER — Total: 40 Marks